Chapter-1

1.Describe computer architecture and computer organization.

Computer Architecture

- Computer architecture includes system attributes visible to programmers.
- These attributes affect the logical execution of a program.
- Examples: instruction set, data representation (numbers, characters), I/O methods, and memory addressing techniques.

Computer Organization

- Computer organization refers to the operational units and their internal connections that realize the architectural specs.
- It includes hardware details not visible to programmers, such as control signals, computer-to-peripheral interfaces, and the type of memory used.

2.Importance of the Distinction

1.This distinction has always been significant, and still is.

2.Many computer families share the same architecture but differ in organization, which is why programmers may see architectural

similarities among different models that have differing hardware implementations.

3.why we need computer architecture and computer organization.

Importance of Computer Architecture

- Defines functionality: It specifies what a system can do—such as its instruction set and the ways it represents and processes data.
- Programmer interface: Architecture provides the features and tools visible to programmers, enabling software development without needing hardware details.

Importance of Computer Organization

- Realizes functionality: Organization focuses on how system features are physically implemented, including connections between components, control signals, and memory technology.
- Optimization: It allows designers to optimize performance, cost, and efficiency by choosing effective ways to implement architectural features (e.g., using special hardware or shared resources)

4.The IBM System/370 architecture

The IBM System/370 architecture, introduced in 1970, allowed customers to use different models with the same architecture, ensuring software compatibility as technology improved. In microcomputers,

architecture and organization are more closely linked, with both evolving together to introduce new features and complexity. Understanding both aspects is crucial for designing efficient and compatible computer systems.

5. What, in general terms, is the distinction between computer structure and computer function?

Structure: Describes how system components are interrelated.

It describes the organization and arrangement of these parts within the system

Function: Describes how each component operates as part of the system

It explains what each component does within the structure of the system.

6. The four main functions of a computer

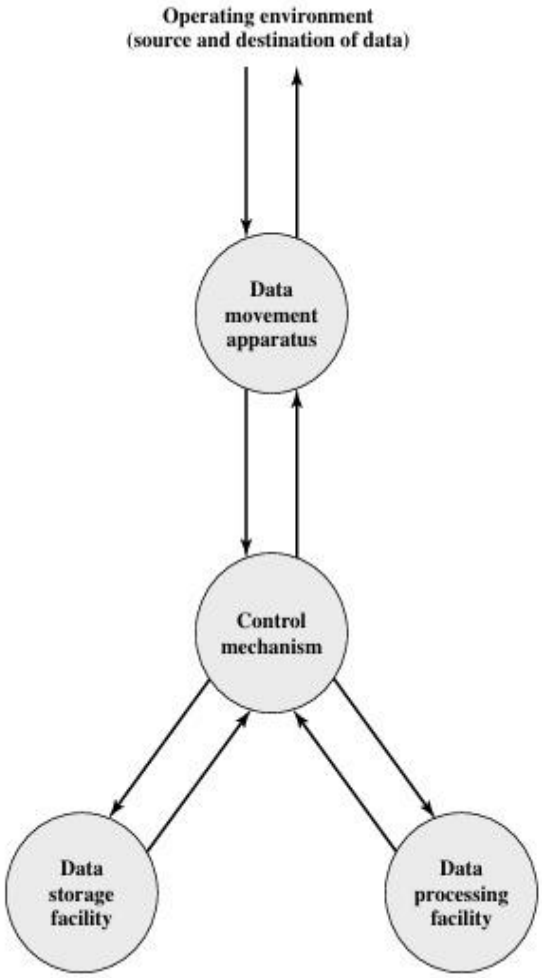
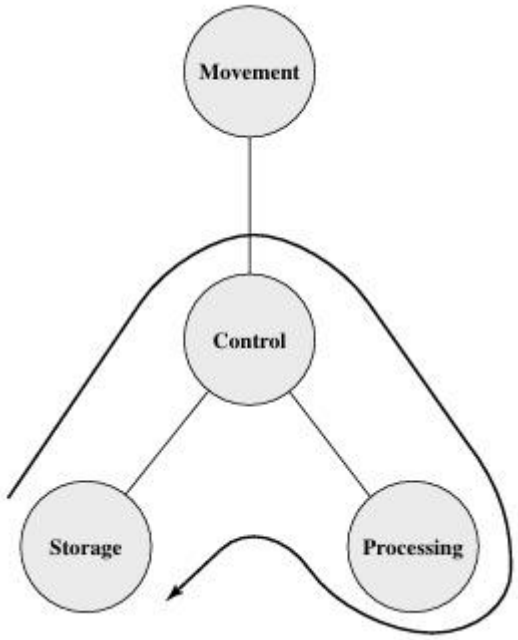


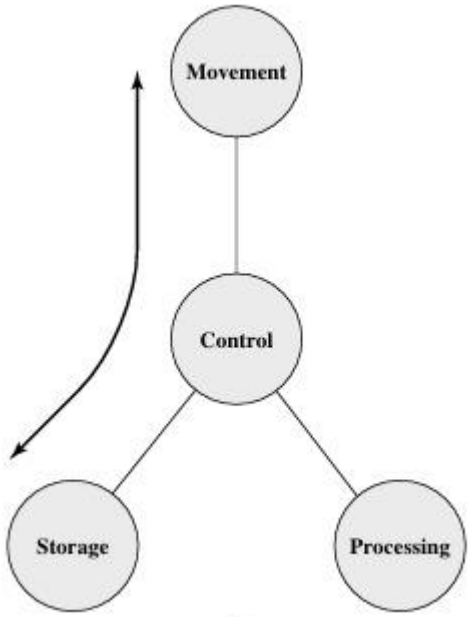
Figure 1.1 A Functional View of the Computer

A computer performs four fundamental functions:

Data processing: Manipulating data in various ways to produce outputs.

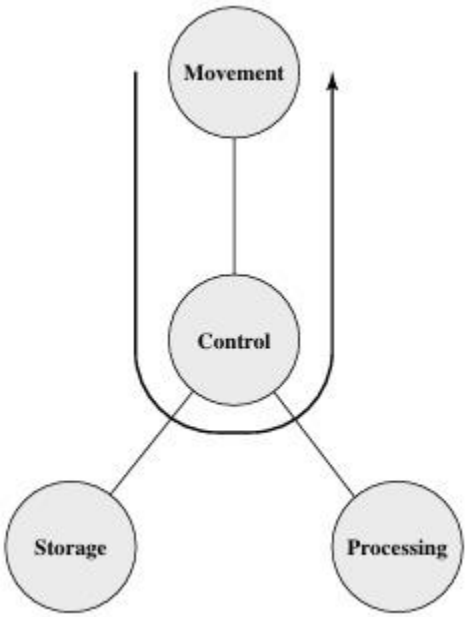


Data storage: Temporarily or permanently storing data for ongoing or future use.



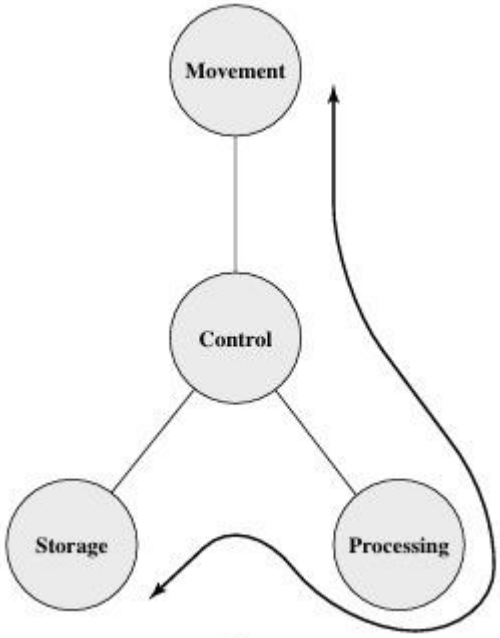
(b)

Data movement: Transferring data within the computer or between the computer and external devices.



(a)

Control: Directing and managing the operations of other components and processes.



7. List and briefly define the main structural components of a computer.

- **Central processing unit (CPU)**: Controls the operation of the computer and
- performs its data processing functions; often simply referred to as processor.
- **Main memory**: Stores data.
- **I/O:** Moves data between the computer and its external environment.
- **System interconnection**: Some mechanism that provides for communication among CPU, main memory, and I/O. A common example of system interconnection is by means of a system bus, consisting of a number of conducting wires to which all the other components attach.

8.List and briefly define the main structural components of a processor.

the most interesting and in some ways the most complex component is the CPU. Its major structural components are as follows:

• **Control unit**: Controls the operation of the CPU and hence the computer

• **Arithmetic and logic unit (ALU**): Performs the computer's data processing

functions

- **Registers:** Provides storage internal to the CPU
- **CPU interconnection**: Some mechanism that provides for communication

among the control unit,ALU, and registers.